

PARAMESHWARAN ANANDHAKUMAR

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SUMMARY

Software engineer with 3 years at Ford building backend services, data pipelines, and full-stack platforms, now completing an MS in Data Science at the University at Buffalo. Hands-on with Python, PyTorch, and FastAPI, with recent work across agentic LLM systems, RAG, and applied NLP. Looking to bring both engineering depth and ML training to machine learning engineering roles.

SKILLS

- **Programming Languages:** Python, Java, SQL, JavaScript, TypeScript, R
- **Machine Learning & AI:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Pandas, Hugging Face, LLMs, RAG, Agentic AI (ReAct, tool-calling), word embeddings, FAISS
- **Backend:** FastAPI, Spring Boot, Node.js, Django, GraphQL, REST APIs
- **Frontend:** React.js, Redux, Next.js
- **Data & Infrastructure:** PostgreSQL, MySQL, MongoDB, SQLite, Redis, Elasticsearch, Apache Airflow, Docker, Kubernetes, AWS (Lambda, EKS, SageMaker), GitHub Actions
- **Testing:** pytest, JUnit, Jest, React Testing Library

EXPERIENCE

University at Buffalo, Institute for AI and Data Science | Software Engineering Intern Jun 2026 - Present

- Built a local semantic search index over 1,821 faculty profiles using GloVe word embeddings, averaging per-profile keyword vectors and ranking results by cosine similarity, removing any dependency on external LLM APIs.
- Built a record-matching pipeline in Node.js and MongoDB that cross-referenced 275 scraped affiliate records against the 1,821-document faculty database, normalizing email identifiers to reach an 83% match rate with zero duplicates.

Ford Motor Company | Software Engineer Jul 2022 - Jul 2025

- Built and shipped a Service Parts After-Sales platform (React.js, Spring Boot, MySQL) with REST APIs for order management, pricing, and delivery, cutting order processing time 55% and handling 500K+ monthly transactions across 11 dealer networks.
- Integrated vendor procurement workflows with internal ERP systems in Node.js and GraphQL, automating order triggers with priority scoring across 10K+ material SKUs; vendor response time dropped 42% and procurement cycles shortened by 3 days.
- Tuned 50+ SQL queries and reworked schemas on a warehouse serving 2M+ daily records, reducing average query runtime 70% for After-Sales reporting dashboards.
- Designed a material demand forecasting and priority scoring platform (FastAPI, React.js, PostgreSQL) using time-series analysis, letting procurement teams predict buy quantities and rank vendors in real time; stockout risk fell 38% across 3 supply chain networks.

RMD Engineering College | AI/ML Software Engineer Jun 2019 - Sep 2020

- Trained a CNN-based brain tumor classifier on MRI images to ~92% validation accuracy, benchmarking custom CNN, AlexNet, and LeNet architectures and improving over baseline by ~20% through architecture and hyperparameter tuning.
- Cut validation loss ~25% with preprocessing and augmentation pipelines (rescaling, zoom, shear, horizontal flips) and deployed the model behind a Django app serving real-time MRI inference under 1 second.

PROJECTS

WearWise - Agentic AI Personal Stylist | [Github](#)

- Built an agentic AI stylist (Python, FastAPI, SQLite) that recommends outfits from a user's own wardrobe, running a ReAct tool-calling loop that searches wardrobe items and retrieves fashion norms via agentic RAG before recommending.
- Grounded every recommendation in cited knowledge-base sources to cut hallucination, integrated Gemini's multimodal model for clothing-photo interpretation, and covered the system with 28 unit and integration tests.

AutoDev - AI Software Engineering System | [Github](#)

- Built a multi-agent system that turns product requirements into implementation plans, architecture, executable code, and pytest test suites, with a self-correction loop that detects failing tests and regenerates fixes without human input.
- Coordinated 5 specialized LLM agents (Planner, Architect, Coder, Tester, Debugger) through a tool-execution layer running pytest, flake8, Docker, and GitPython; deployed via Docker Compose with FastAPI, PostgreSQL state persistence, and FAISS vector memory.

EDUCATION

University At Buffalo, New York Aug 2025 - Dec 2026

Master of Science, Data Science

Anna University, Chennai India Aug 2018 - May 2022

BTech, Information Technology

HONORS & AWARDS

- Google Professional Machine Learning Engineer | Google | 2026
- Advanced Data Analytics Certificate | Google | 2025
- Top Achiever Award | Ford Motor Company | 2022, 2024
- Smart India Hackathon | Intercollege Hackathon - Winner | 2020